https://github.com/sunnybutrainy/485_Group1

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Accessibility in Software Development

ACM Code of Ethics Case Study

Group 1

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Case Study Chosen and Why We Selected It

We selected this case because accessibility represents a critical intersection between ethnics, technology, and social responsibility. We have a professional and moral duty to ensure our work benefits all users, including those with disabilities.

A powerful example of this intersection is Elon Musk's Neuralink, a brain computer interface designed to restore mobility and communication for people with paralysis. Its potential to transform accessibility is extraordinary, offering individuals with severe disabilities the ability to control devices, type, or move robotic limbs using only their thoughts. This neuroscience technology especially includes users with physical disability, such as paralysis, who have historically been excluded from digital participation.

However, Neurallink also raises serious ethnic and safety concerns. Implementing a chip in the human brain introduces risks such as data privacy violation, long-term neurological effects, and transparency. The data ownership and consent for brain recorded information complicates the issue.

The ACM Code of Ethics emphasizes principles such as "Avoid Harm," "Be fair and take action not to discriminate," and "Contribute to society and to human well-being." These guidelines remind us that while innovation like Neuralink can improve lives, developers must anticipate potential harms and protect users before pursuing commercial or competitive gain.

Our case highlights a real-world dilemma: balancing business pressure, expectations, and user accessibility requirements. It allows players to experience the pressure of choosing between technological ambition and social responsibility, reinforcing how ethical reflection and accountability should remain central in the pursuit of innovation.

Prototype Workflow

The prototype, built in Twine (Harlowe 3.3.9), follows a branching path that immerses the player in the role of a software design team leader at a company developing a web-tool. Team discovers accessibility issues that violate WCAG (Web Content Accessibility Guidelines), while shareholders push for an immediate release. Players must choose how to respond under time and business pressure.

Paths

Intro → Common Scene

"The design team encounters the lack of key accessibility requirements; pressured by company to release by shareholders anyway against WCAG's wishes."

Choices:

Ignore pressure, aim to implement accessibility options → Ethical Choice

Listen to the pressure, ignore implementing → Unethical Choice

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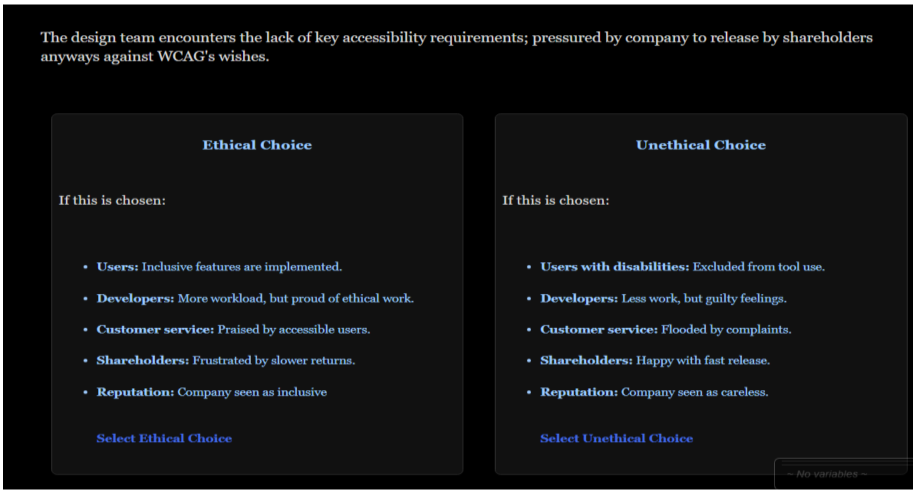

Image 1. First Choice for User to select

Ethical Path (⟶ "Ethical Choice")

The player chooses to resist pressure and ensure accessibility compliance.

The company prioritizes its users (⟶ EC1)

The company listens to shareholders (⟶ EC2)

Outcomes:

EC1: Users are satisfied, shareholders are upset, but accessibility goals are achieved.

"The web-tool gets the accessibility it needs… the company admits such a delay was necessary."

EC2: Company gives in temporarily, leading back toward "Unethical Choice" as a reflection of ethical relapse.

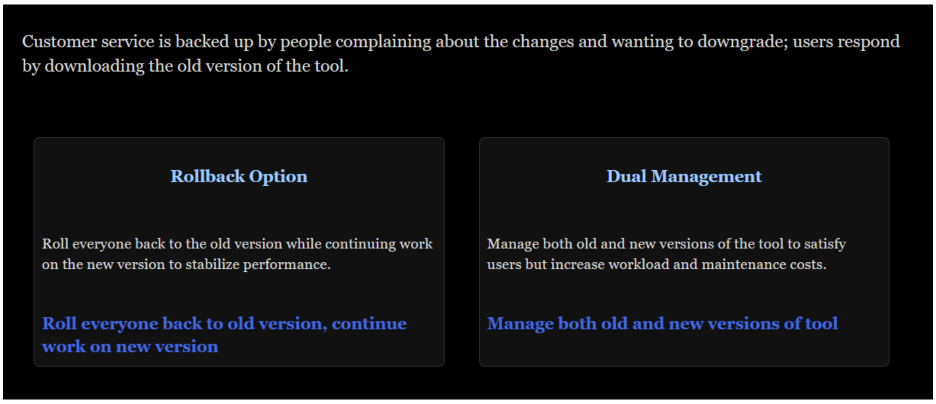

Image 2. Selections for Unethical Path

Unethical Path (⟶ "Unethical Choice")

The player yields to shareholder pressure and neglects accessibility.

Roll back to old version (UEC1): The team manages user backlash by reverting while continuing development.

Outcome ⟶ Accessibility achieved but at higher operational cost.

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Manage both versions (UEC2): Company maintains both old and new tools.

Outcome → Both shareholders and users become dissatisfied, damaging reputation.

Dilemmas players experience

User Accessibility vs. Business Deadlines

- Should the development team delay release to comply with accessibility standards or meet investor expectations?
- ACM Principle: Contribute to society and human well-being; avoid harm

Profit vs. Inclusion

- Management pressures developers to prioritize revenue and speed over inclusivity.
- Equity vs. efficiency whether to serve the majority or protect marginalized users.

Short-Term Gains vs. Long-Term Trust

- Ignoring accessibility may increase immediate profits but risks long-term loss of user trust and brand reputation.
- ACM Principle: Be honest and trustworthy; respect privacy and the public good.

Team Morale vs. Corporate Pressure

- Team members face internal conflict, follow moral conviction or obey superiors demanding release.
- Courage to act ethically in hierarchical environments.

Responsibility Distribution

- Who is accountable for accessibility: developers, designers, or executives?
- ACM Principle: Uphold professional responsibility and leadership in ethical conduct.

Transparency with Stakeholders

- Should the company disclose to users that the tool does not meet accessibility standards?
- Honesty vs. protecting corporate image.
- ACM Principles: Uphold professional responsibility and leadership in ethical conduct.

External References

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